

Skills Summary

When a Number Rule Stops Working

Use this reference while completing your worksheet or when studying.

Skill 1 — Testing a rule instead of trusting it

Rule: A claim that starts with *every*, *all* or *always* is only safe until one case breaks it. A few examples that agree prove nothing. One example that disagrees — a **counterexample** — settles it.

So: keep generating terms past the ones you were shown, and check each against the claim.

Example: A pattern starts at 5 and adds 6 each time. Someone looks at the first few terms and claims every term is prime. Find the first term that is not prime.

Step 1. The claim says every term, so I keep listing terms and test each one for factors instead of stopping where the examples stopped.

Step 2. The terms are 5, 11, 17, 23, 29, then the 6th term: $5 + 5 \times 6 = 35$, and $35 = 5 \times 7$, so it is not prime. First term that breaks the claim: **35**

Try it: A pattern starts at 2 and adds 3 each time. Find the first term that is not prime.

8 :pɛk

Skill 2 — Last digits do not decide primes

Rule: A number is prime when its only factors are 1 and itself. The last digit cannot tell you that.

The last digit rules out some primes (a number ending in 0, 2, 4, 5, 6 or 8 above 5 is never prime), but it can never guarantee one. To settle it, hunt for a factor pair.

Example: Someone claims every number ending in 3 is prime. Find the smallest counterexample greater than 3.

Step 1. To break an “every” claim I need one number ending in 3 with a factor pair other than 1 and itself, so I test them in order.

Step 2. 13 is prime and 23 is prime, but the next one splits: $3 \times 11 = 33$. Smallest counterexample: **33**

Try it: Someone claims every number ending in 9 is prime. Find the smallest counterexample greater than 9.

68 :pɛk

Skill 3 — Repairing a pattern that drifted

Rule: In an add-the-same-number pattern, every step must be identical.

term $n = \text{first term} + (\text{step size}) \times (n - 1)$

Check a written list by subtracting neighbours: the differences must all match. The first place they do not is where the mistake is, and every term after it inherits the error.

Example: A pattern starts at 3 and adds 5 each time. Someone wrote 3, 8, 13, 20. Find the correct 4th term.

Step 1. The differences in the written list are 5, 5, then 7, so the list drifted at the last step — I rebuild that term from the rule rather than from the wrong list.

Step 2. The 4th term is 3 steps past the start: $3 + 3 \times 5 = 18$. Correct 4th term: **18**

Try it: A pattern starts at 7 and adds 4 each time. Someone wrote 7, 11, 15, 20. Find the correct 4th term.

check: 18

(!) Watch out: Terms that share a property — all odd, all prime so far, all two-digit — do not tell you the rule that made them. Ask what is *done* to get from one term to the next, then test that on a term nobody showed you.